

Earthquake's Magnitude Estimation from Early P-Wave Observations Using Machine Learning

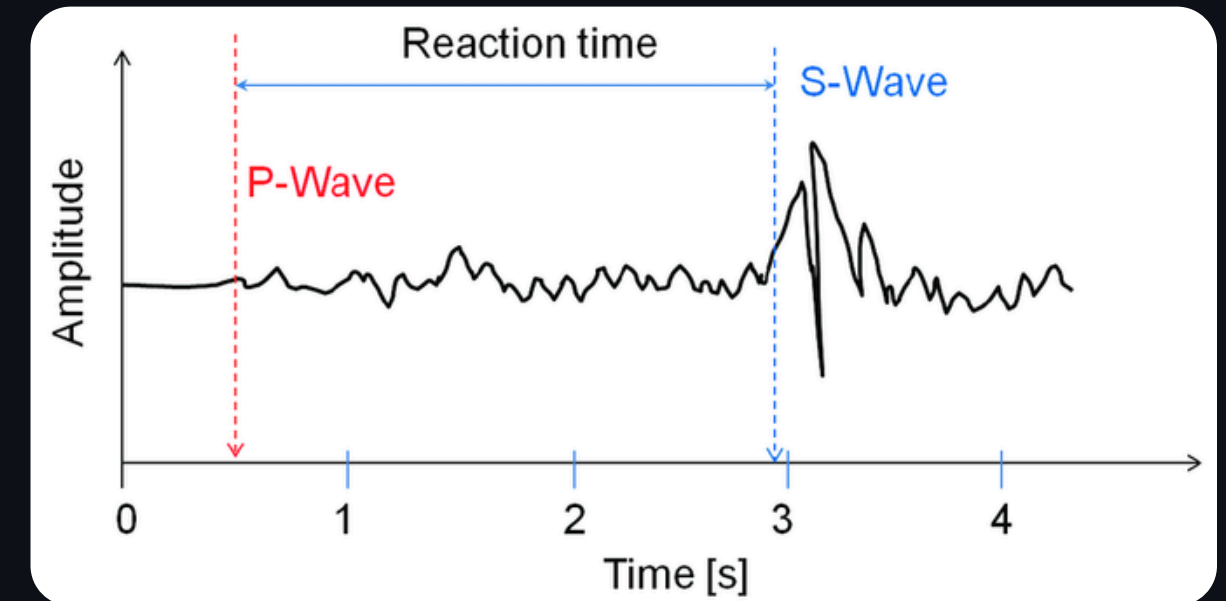
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Problem Statement & Aim of Study

The Core Question

To minimize the damage caused by earthquakes, **Earthquake Early Warning (EEW)** systems are utilized. Our aim is then to investigate how well the relationship between the first observations of an earthquake and its subsequent magnitude can be learned by a machine learning model.



M5.2

Seismic Terminology

P-Waves (Primary/Pressure Waves)

- Fast
- Moderate Energy Carried

S-Waves (Secondary/Shear Waves)

- Slow
- Significant energy propagates

S-P Time Difference → Distance Estimation

- Gap between P-wave and S-wave arrivals.
- Seismograph measurements: velocity in m/s.

Dataset: INSTANCE

~1.2M

Raw instances

~102k

After undersampling

50,000

M 1.0–3.0

50,000

M 3.0–5.0+

PREPROCESSING PIPELINE

1. **Data Source:** Italian seismic records. Velocity waveforms in m/s, m/s².
2. **Filtering:** Earthquakes recorded in velocity (m/s) & manually picked P-waves arrivals
3. **Sampling:** 50,000 for M1.0–3.0, 50,000 for M3.0–5.0, ~1.7k M5.0>
4. **Preprocessing:** Each waveform normalised to the range $[-1, 1]$.

Mathematical Foundation - Short-Time Fourier Transform

STFT Formula

$$X[m, k] = \left| \sum_{n=0}^{N-1} x[mH + n] \cdot w[n] \cdot e^{-j2\pi kn/N} \right|^p$$

Hann window function

$$w[n] = 0.5 \left(1 - \cos\left(\frac{2\pi n}{L-1}\right) \right), \quad n = 0, 1, \dots, L-1$$

$x[mH + n]$ - input seismic waveform

$w[n]$ - Hann window function

H - hop length (samples)

N - FFT size

m - frame index

k - frequency bin index

p - power exponent

Mathematical Foundation - Model Architecture

Convolutional Neural Networks

Convolutional Neural Networks

$$(X * K)[i, j] = \sum_{m=0} \sum_{n=0} X[i + m, j + n] \cdot K[m, n]$$

MLP

Multilayer Perceptron

$$\hat{y} = W_n \cdot \sigma(\dots \sigma(W_2 \cdot \sigma(W_1 \cdot x + b_1) + b_2) \dots) + b_n$$

LSTM

Long Short-Term Memory

$$\begin{aligned} f_t &= \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \\ i_t &= \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \\ o_t &= \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \end{aligned}$$

Mathematical Foundation - Loss Function

Loss Function

Huber Loss

$$L_{\delta}(y_i, \hat{y}_i) = \begin{cases} \frac{1}{2} (y_i - \hat{y}_i)^2 & \text{if } |y_i - \hat{y}_i| \leq \delta \\ \delta \cdot (|y_i - \hat{y}_i| - \frac{1}{2} \delta) & \text{otherwise} \end{cases}$$

Evaluation Metrics

Mean Residual

$$\text{Bias} = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)$$

Mean Absolute Error

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

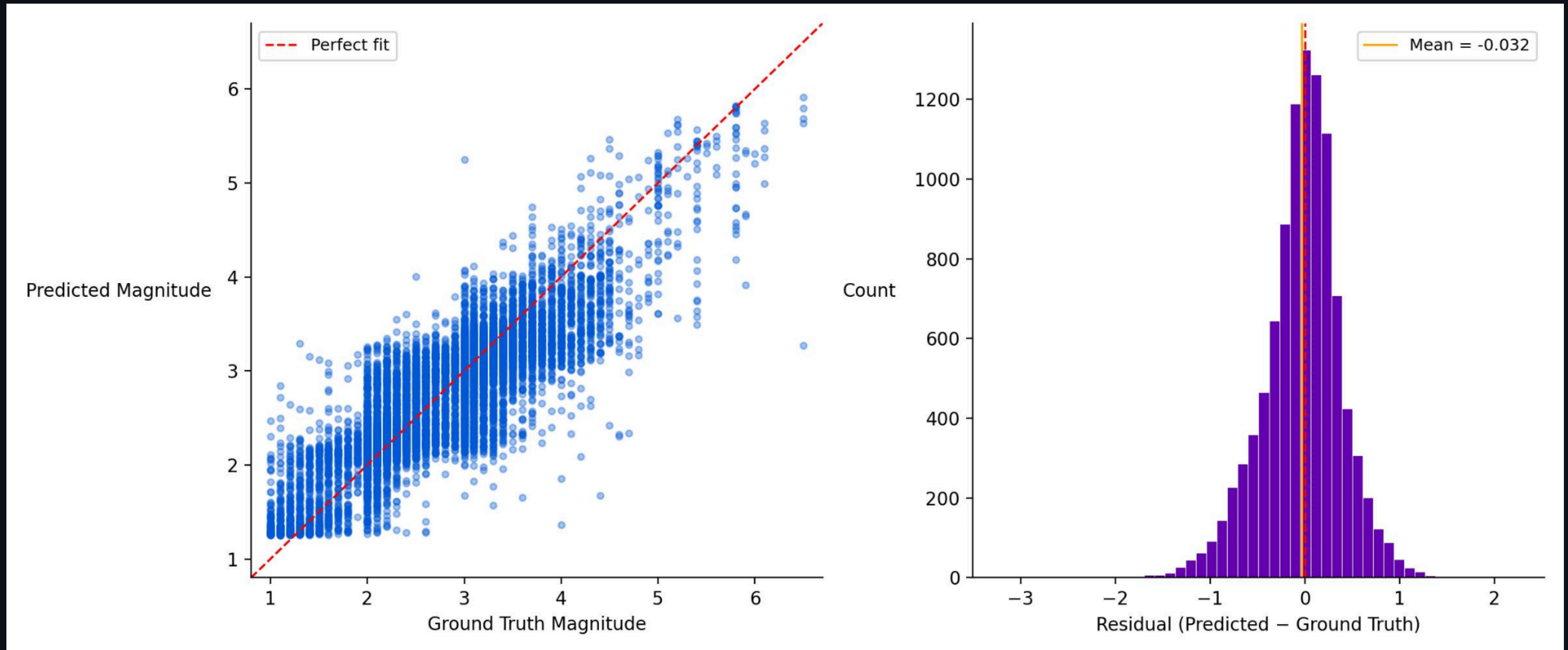
Spearman's correlation

$$\rho = \frac{\sum_{i=1}^N (r_i - \bar{r})(s_i - \bar{s})}{\sqrt{\sum_{i=1}^N (r_i - \bar{r})^2 \cdot \sum_{i=1}^N (s_i - \bar{s})^2}}$$

Coefficient of Determination

$$R^2 = 1 - \frac{\sum_i^N (y_i - \hat{y}_i)^2}{\sum_i^N (y_i - \bar{y})^2}$$

Experiments

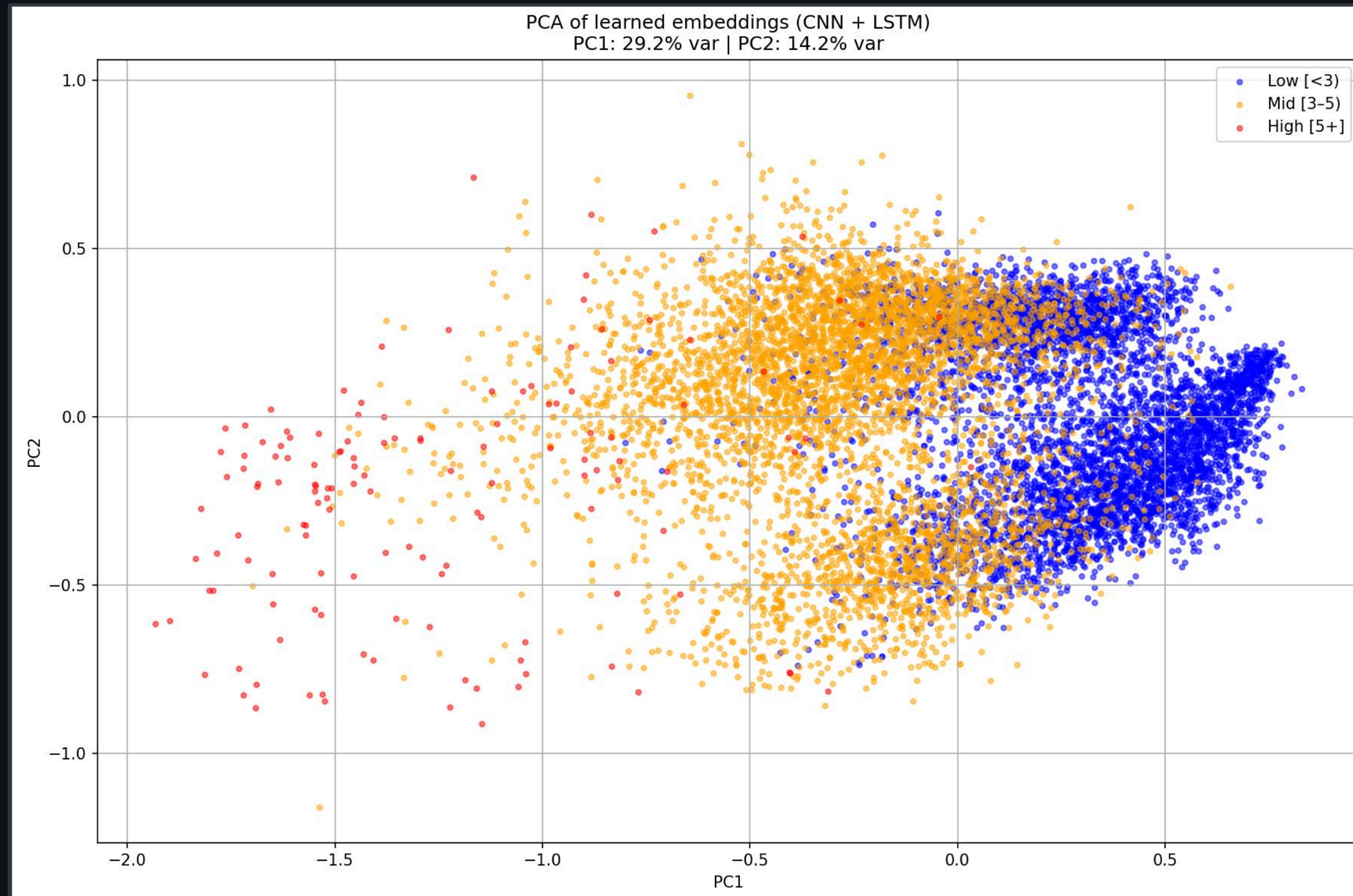


Experiments: Quantitative Results

CONFIGURATION:
learning rate = 0.001368
weight decay = 0.005068
batch size = 128
scheduler = cosine
eta min = 8.6534e-6

Time Window (in seconds)	Bias	MAE	Spearman's Corr	R ²
3	-0.040	0.3699	0.7755	0.6483
5	-0.071	0.3453	0.8028	0.6851
8	-0.027	0.3217	0.829	0.7225
10	-0.030	0.3114	0.8392	0.7478

Learnt Representation – Principal Component Analysis



Conclusion

Revisiting the Aim

We set our aim to investigate how well the relationship between the first signals of an earthquake and its subsequent magnitude can be learned. The results show the proposed method manages to capture the relationship with a meaningful, but bounded error.